RENDERING THE INVISIBLE: DIGITAL AND ORAL INTERPRETATION OF DIFFICULT HISTORY AT THE UNIVERSITY OF VIRGINIA'S ANATOMICAL THEATRE

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ABSTRACT

This preliminary qualitative study examines how visitors engage with digital and oral interpretations of difficult histories at the University of Virginia's Anatomical Theatre, a site whose physical structure was demolished in 1939 and whose history of enslaved labor and grave robbing remains largely unknown to the campus community. Digital initiatives addressing dark and difficult heritage remain understudied, with little research on how visitors actually experience this kind of content. Fifteen participants took part in an introductory qualitative study comparing an oral tour paired with an augmented reality (AR) web application and 3D reconstruction to the same oral tour paired only with the site's existing ground plaque. The results, while preliminary, point toward directions for future research on both application design and methodology. Most participants strongly preferred the oral portion of the tour and treated the AR app and 3D model as a complementary visual reference rather than a replacement for a human guide. This raises an open question about the cognitive and emotional difference between a person sharing difficult history in person and a fully immersive digital experience that attempts to replicate that connection. A secondary finding was that participants in the AR group engaged more with the space in physical and spatial terms and used the app as a visual reference, while participants who received only the ground plaque left with more confusion about what an anatomical theatre actually was and made fewer observations about the building's physical scale and presence.

KEYWORDS

difficult heritage; augmented reality; oral history; digital heritage interpretation; qualitative methods

1. INTRODUCTION

Buildings preserve histories that written records omit. When buildings disappear, or even just change in function and temporal context, the histories they contained become abstract. The University of Virginia's Anatomical Theatre exemplifies this erasure.

Designed by Thomas Jefferson but constructed after his death in 1826 by enslaved laborers, including the stonemason Peyton Skipwith (Encyclopedia Virginia, 2024; Puzio, 2015), the building served as UVA's site of medical education for over a century. Inside, medical students performed dissections while an enslaved man named Lewis, derogatorily nicknamed "Anatomical Lewis" by students, prepared and disposed of the bodies from 1839 to 1857 (Puzio, 2015; Memorial to Enslaved Laborers, n.d.). Many cadavers were stolen from African American burial grounds in Charlottesville, as the demand for anatomical subjects exceeded the legal supply (Gates, 2017). The building was demolished in 1939 during construction of Alderman Library (Gates, 2017; Puzio, 2015). Today, a small ground plaque marks the site, but most students do not know it exists.

Difficult heritage sites are those that invoke remembrance of events that were cruel, atrocious, and awkward for public reconciliation (Macdonald, 2009; Dima et al., 2024). There are many foundational works examining the use of AR/VR and 3D reconstruction for general learning purposes (e.g., Dede et al., 1996; Johnson et al., 1999; Roussou et al., 2006) and for lost or hard-to-access historical sites (Singh et al., 2015). There is less work investigating these same technologies' impact at difficult heritage sites specifically. Human emotions are connected to our memory and learning outcomes (Immordino-Yang & Damasio, 2007), and AR and 3D technologies offer a more immersive experience that has been recognized for its ability to accommodate the heavy emotions that accompany these histories and to shape both the perception of history and learning outcomes.

In the Atlas of Lost Rooms project, researchers built a 3D reconstruction of the Magdalene Laundry in Ireland, incorporating the narratives of women who were imprisoned there (Hamill, 2021). This is a remote learning experience with preliminary results on user experience. Pickard (2022) examines the literature and case studies of representing difficult histories in museum and heritage settings and characterizes visitor emotions and responses to these interpretive experiences; her work does not compare these results to the emotional effect of an oral tour. Many universities have used AR/VR applications to highlight the histories of the people they enslaved without disrupting or altering campus architecture, and to explore interactive experiences more broadly (Dowell & Baynes, n.d.; Gladney et al., 2024). UVA itself has developed a self-guided digital walking tour of its history of enslaved labor (University of Virginia Library, 2020), and the present study builds on this foundation with its own AR web application focused on a single site. Many of these are project outcomes and do not examine how these digital approaches compare to traditional tours or how they affect perceptions of campus history.

This preliminary study builds on this existing work to examine the comparative effect of 3D and AR technologies on in-person tours at the difficult history site of the University of Virginia's Anatomical Theatre.

2. METHODOLOGY

The 3D models of the theatre were generated in Python code using the geometry from the Historic Americans Buildings Survey (HABS) drawings of Jefferson's Anatomical Theatre. Photos and historical descriptions of the site informed deviations from this design. The models were embedded into an easy to access web application. Fifteen participants were recruited for the study. The participants varied in age range with younger participants. making the majority of those who reported their age: 3 were aged 55-64, 2 45-54, 1 25-34, and the remaining 8 were 18-24. Of participants who reported affiliation with UVA: 5 participants were students, 1 was an employee of the school, 2 were alumni, and 4 had no affiliation. The participants racial identity and family history in the United States was also collected. All fifteen completed the survey, and fourteen completed the interviews. Participants were assigned to experience either the AR web app or the existing ground plaque. In multiple sessions run in groups of two to three, all participants received the same oral tour, accompanied by either the 3D reconstruction and AR experience or just the ground plaque. After the brief tour, participants filled out a survey covering their emotional state, their perception of UVA history, and optional demographics. They then sat for a semi-structured interview that expanded on their survey answers and, for the AR group, included questions about the usability and design of the web app. The transcript of these interviews were read line by line and coded across 16 themes.

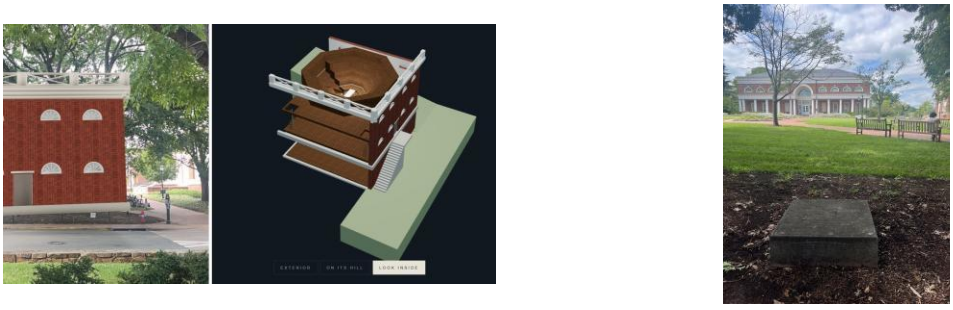


Figure 1. The AR web app's augmented-reality overlay and interactive 3D model (left) and the ground plaque marking the site (right).

3. FINDINGS

3.1 Survey Results

The small sample size of the survey cannot support any conclusions or significant correlations between tour condition and responses. The survey was largely ineffective at capturing perceptual differences, with most group means clustering between 4.0 and 5.0 (see Table 1). The ground plaque condition had higher scores overall for historical presence and perception of UVA history. The two exceptions favoring the AR condition were Q1 (imagining appearance, +0.08) and Q6 (felt emotional, +0.25), both very small differences. Q7 asked participants to select the emotions they experienced; responses were as follows: curious, 13/15; moved, 9/15; sad, 8/15; surprised, 6/15; uncomfortable, 2/15; indifferent, 0/15. The survey did not distinguish experiences between the two conditions well. Hearing the information through an oral tour seems to have emotionally oriented most participants enough to motivate strong agreement on the Likert-scale rankings of their experience.

Table 1. Mean survey ratings by condition (5-point scale)

Measure	AR Web App (n = 8)	Ground Plaque (n = 7)
Q1 — Could imagine past appearance	4.50	4.43
Q2 — Felt connected to past people	3.62	4.29
Q3 — History felt real	4.62	4.71
Q4 — Past felt present	4.00	4.14
Q5 — Could picture activities happening	4.25	4.86
Q6 — Felt emotional	4.25	4.00
Q9 — More aware of UVA's history	4.38	4.86
Q10 — More connected to UVA's history	4.00	4.43
Q11 — Changed thinking about the site	4.38	4.57
Q12 — Would recommend	4.75	5.00

3.2 Interview Results

3.2.1 Oral Tour Preferred, App Seen as Complementary

A primary theme across participants, which corroborates the survey results, was a strong preference for the oral component of the tour. This theme was coded eleven times across both conditions. Participants generally agreed that having a human tour guide enabled another level of interaction and heightened their feelings of connection. One participant in the ground plaque condition noted that "having someone tell the history makes it feel a little more real... helps me connect to the people who were affected." Another participant, who experienced the web app, described it this way: "you have to have the oral. It's like affirmations, it brings the information to life."

Participants who experienced the web app also valued the digital experience; one said, "I think they both work together, I wouldn't want either one separately." Overall, the digital experience was coded primarily as a complementary visual reference to the tour. Participants described it as suiting their status as visual learners and said they referred to it during the tour to orient themselves with the information. When asked whether the application should be made more immersive, many participants noted that adding more elements to the web app could take away from the tour experience; one participant negatively noted the possibility of "a lot of text to sit there and read," and another compared the suggestion to museum tours: "I have been to museums where you're trying to read the plaque at the same time the guide is giving you information... competing interests there." A few participants did say that additional text labels or human figures and props could improve engagement.

3.2.2 Group Differences in Language and Understanding

The web app group used more visceral and imaginative language, coded in six of seven participants. One participant said, "it's like I was there, I could see the people in their suits walking into the classroom," and another said, "the AR experience is like, 'this was a real building.'" In addition, responses from the AR group involved a physical and spatial reckoning with the history. This code described questions or comments relating to participants' perception of the building's physical embodiment. Participants referred to the size of the building, its proximity to the Academical Village, and practical questions that arise naturally from this, such as smell, waste removal, and student perceptions of the theatre.

This spatial understanding stands in contrast to the coding for participants who experienced the ground plaque. Although they were highly engaged and found the tour informative, they had difficulty forming a baseline understanding of what an anatomical theatre actually is and how the building interacted with its

environment. One participant described it this way: “there’d be different buildings, different areas leading up to that grave site...” — a warped mental model of the anatomical theatre’s processes and structure. Another participant, asked what an anatomical theatre was, said, “what I had in mind is just, like, a pure stone box... I have no clue what an anatomical theatre is.” These participants retained the historical narrative of grave robbing and its racial dimensions, but not a basic sense of what the building actually was or looked like.

3.2.3 Group Differences in Language and Understanding

Both groups expressed relief that the building no longer exists, recognized the contrast between the importance of medical research and the harm done to Black communities, commented on the lack of visibility of this history and how easy it is to overlook the ground plaque, and were curious to know more.

4. DISCUSSION

It is important for institutions to develop effective ways of teaching sensitive histories and to encourage approaches that foster learning through emotional connection. Managing the display of difficult histories is a challenge shared by many universities in the United States and abroad, and it extends beyond the institution of slavery. This preliminary study points toward examining the relationship between an oral guide and digital history tools. Many 3D reconstruction tools and AR tours are self-guided and designed to be individually immersive at scale.

Participants’ strong valuation of the oral portion of the tour suggests a relationship between digital and oral histories that warrants further examination. One possible model is a simplified 3D/AR app for in-person tours, kept as a minimal visual aid, developed alongside a more immersive standalone experience with text and pre-recorded audio. Multimodal approaches are generally considered advantageous for learning (Mayer, 2009), though there may be something unique about in-person sharing specifically for processing difficult histories. Future studies could build on this by comparing an in-person tour with a human guide plus a digital reference to a fully digital tour with pre-recorded audio.

It is also possible that the use of a visual aid caused split attention amongst the participants and that visual aids can cause users to engage less with the human narrative. This is also something to be explored in future work investigating the effects of tours on visitors.

5. LIMITATIONS

This preliminary study is limited by its small, uneven sample size of participants and interviewees. It is also limited by its uneven comparison across two mediums (plaque and AR) that differ starkly. Since many participants valued the web app primarily as a visual reference, it is unclear whether their experience would have been similar had they been given an image or diagram rather than the 3D reconstruction. The survey questions need revision to better distinguish differences in experience among participants. Different aspects of the web app (AR, interactive 3D model) could be tested separately to avoid confounding results. This report is meant to identify next research steps in AR app design, tour design, approaches to measuring participants’ cognitive and perceptual experience, and improvements to both the experimental methodology and the digital application itself.

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